Argument 6: Extrapolation from small models suggests induction heads are responsible for the majority of in-context learning in large models.

STRENGTH OF ARGUMENT FOR SUB-CLAIMS

	Small Attention-Only	Small with MLPs	Large Models
Contributes Some			Analogy
Contributes Majority			Analogy

This argument is really an extended inference from all the above arguments. Arguments 1-5 present fairly strong evidence that in small transformers (especially small attention-only models), induction heads are responsible for the majority of in-context learning, while for large transformers, the evidence is not as strong. To what extent can one reasonably infer from small models that the same thing is happening in larger models? Obviously this is a matter of judgment.

The measurements in the model analysis table look fully analogous between the small attention-only, small models with MLPs, and full-scale model cases. Provided there's more than one layer, all of them go through a phase change. All of them have the same sharp increase in in-context learning, with the same rough amounts before and after the transition. All of them trace similar paths in PCA space. All of them form induction heads.

If things change from the small model case to the large model case, where do they change? And why is there no visible sign of the change in all our measurements?

On the flip side, there are many cases where large models behave very differently than small models (see discussion of phase changes with respect to model size in Related Work). Extrapolating from small models to models many orders of magnitude larger is something one should do with caution.

The most compelling alternative possibility we see is that other composition mechanisms may also form during the phase change. Larger models have more heads, which gives them more capacity for other interesting Q-composition and K-composition mechanisms that small models can't afford to express. If all "composition heads" form simultaneously during the phase change, then it's possible that above some size, non-induction composition heads could together account for more of the phase change and in-context learning improvement than induction heads do.

Model Analysis Table

The arguments above are based on analysis of 34 decoder-only Transformer language models, with different snapshots saved over the course of training, for one run of training per model. The models are drawn from four different model series as follows:

- "*Small, attention-only models*", a series of models (from 1-layer to 6-layer) that do not have MLPs, and were trained specifically for the present investigations.
- "*Small models with MLPs*", a series of models (from 1-layer to 6-layer) that have both attention and MLP layers, and were trained specifically for the present investigations.
- *"Full-scale models",* a series of successively-larger models with MLPs (ranging from 4 layers and 13M parameters, up to 40 layers and 13B parameters), that are used as the basis for multiple projects at Anthropic.
- "*Smeared key models",* a targeted architectural experiment designed to allow transformers of any depth to express induction heads.

The dataset used for training the small models and smeared key models was an earlier version of the dataset described in Askell *et al.* [18] , consisting of filtered common crawl data [19] and internet books, along with several other smaller distributions [20] , including approximately 10% python code. The full-scale models were trained on an improved version of roughly the same data distribution. Additionally, another set of the small models was trained on a different dataset, consisting of just internet books, to explore the impact of varying the dataset. All models trained on a given dataset saw the same examples in the same order. Models never saw the same training data twice.

For more details on model architecture and training, continue past the table below, to the Model Details section.

In the Model Analysis Table below, each row includes a brief summary of the measurement shown. For a more in-depth explanation of the data collection and results analysis, see the Appendix.

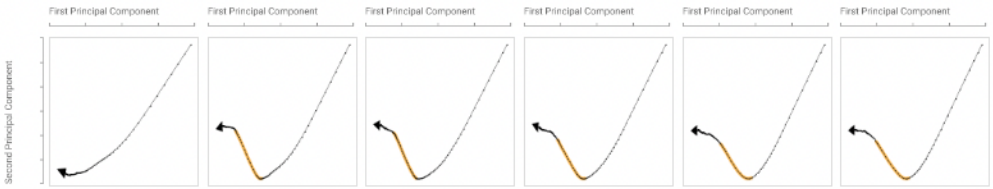
SMALL ATTENTION-ONLY TRANSFORMERS

This experiment series studies small *attention-only* transformers, because these are the models we have the strongest theoretical understanding of (see the framework paper). For these small models, we're able to capture snapshots with a high time resolution over early training. These experiments are the ones that most cleanly demonstrate the phase change and link it to induction heads. Note how (1) many models undergo a discontinuous phase change (seen most clearly in the PCA plot and in-context learning score); (2) it appears to be caused by induction heads (see ablations); and (3) the only model without a phase change is the one-layer model. More...

ONE LAYER (ATTENTION-ONLY) | TWO LAYER (ATTENTION-ONLY) | THREE LAYER (ATTENTION-ONLY) | FOUR LAYER (ATTENTION-ONLY) | FIVE LAYER (ATTENTION-ONLY) | SIX LAYER (ATTENTION-ONLY)

PCA OF TOKEN LOSSES

The vector of per-token losses is a way to map different neural network behavior to the same vector space. We take 10,000 individual token predictions per model, and project them onto the first two principal components. This shows how the large-scale-behavior of multiple networks evolve over training. More...

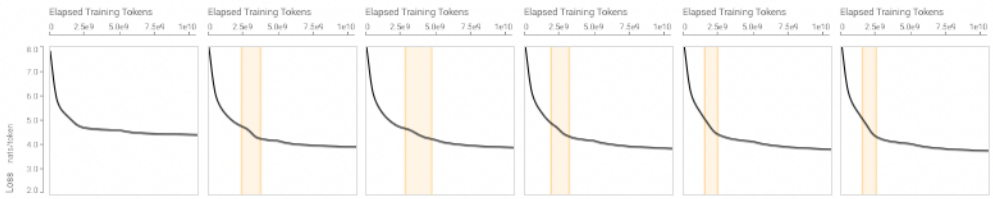


MODELS OVER TRAINING

LOSS OVER TRAINING

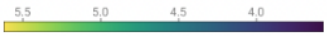
Loss curve showing $\mathrm{Loss}(n_{\mathrm{train}})$ after the model is trained on n_{train} tokens of training data. More...

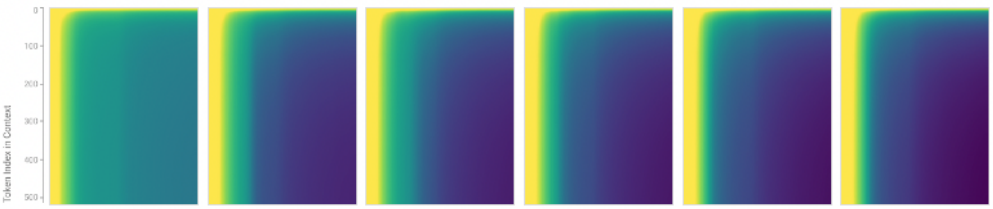
The orange band denotes phase change interval.



LOSS OVER TRAINING, BROKEN DOWN BY CONTEXT INDEX

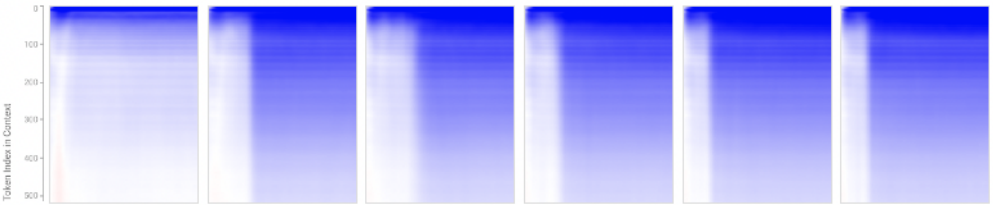
Heatmap of $\mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}})$, the average loss of i_{ctx} token in context after n_{train} elapsed tokens of training. More...



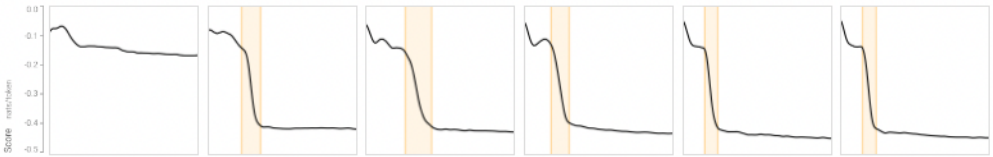


DERIVATIVE OF LOSS WRT CONTEXT INDEX

Heatmap of $d\mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}})\ /\ d\ln(i_{\mathrm{ctx}})$. The log vertical partial derivative of the above graph. More...



"IN-CONTEXT LEARNING SCORE"

$\mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}} = 500)$
$- \mathrm{Loss}(n_{\mathrm{train}},\ i_{\mathrm{ctx}} = 50))$

How much better the model is at predicting the 500th token than the 50th? We use this as a proxy for how good the model is at in-context learning. More...